Cloud Computing

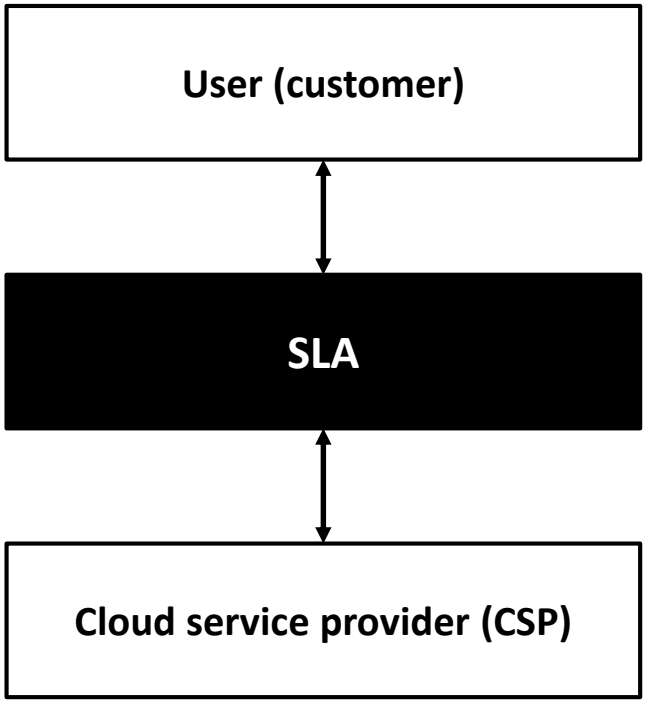
SLAs, cost models, data protection law, migration and multi-cloud

SLA (Service Level Agreement)

A formal contract between a cloud service provider (CSP) and the customer that defines the level of service expected from the provider.

Why is an SLA used?

1. Clarity of expectations
2. Reliability assurance
3. Accountability
4. Customer trust
5. Business continuity



Key components of an SLA

1. **Service availability**
   Uptime
2. **Performance metrics**
   Application response times, data transfer speed, API responsiveness
3. **Support and response times**
4. **Data security and privacy**
   Encryption and compliance standards
5. **Data backup and recovery**
6. **Penalties for non-compliance**
7. **Termination terms**

Consumption-based model

CapEx Capital expenditure	OpEx Operational expenditure	Consumption-based Pay for what you use
Upfront investment by an organization to purchase, upgrade or maintain physical assets like hardware, servers and data centers.	Ongoing costs of running and maintaining services, often subscriptions or pay-as-you-go models.	Customers pay only for the cloud resources they use, like utility charges for electricity or water.

Digital Personal Data Protection Act, 2023

Important terms

1	**Digital data**	Data in a form suitable for being communicated, interpreted or processed
2	**Data processing**	Automated operations on digital personal data, such as collection, storage, use and sharing
3	**Personal data**	Any data about an individual who is identifiable from that data
4	**Data Principal**	The individual to whom the personal data relates
5	**Data Fiduciary**	The person or entity that decides the purpose and means of processing personal data

Seven-step model of cloud migration

#	Step	What happens
1	**Assess the environment**	Evaluate legacy systems; find workloads fit for the cloud
2	**Define the business case**	Improve scalability, security, compliance; cut costs
3	**Select the provider**	Choose AWS: security, scalability, finance-specific services
4	**Plan the migration**	Phased plan: non-critical workloads first, core banking last
5	**Migrate the workloads**	AWS DMS for databases, Snowball for secure data transfer
6	**Validate the migration**	Rigorous testing to meet regulations like PCI DSS
7	**Optimize and modernize**	Go serverless and cloud-native to improve customer experience

Challenges in cloud migration

Challenge	Why it happens
Data security and compliance risks	Sensitive data may be exposed during transfer
Downtime and disruption	Migration can interrupt services
Cost management	Hidden costs in storage, transfer and tools
Application compatibility	Legacy applications may not work in the cloud
Data loss risks	Data may be lost or corrupted during transfer
Vendor lock-in	Dependency on a single provider's ecosystem
Skill gaps in cloud technologies	Teams may lack expertise
Performance optimization	Resources may not be optimized post-migration

Multi-cloud

Using services from multiple cloud providers in a single environment.

Goals

- **Avoid vendor lock-in**
- **Improve redundancy and resilience**
- **Optimize for the best services each provider offers**